

DDPG vs DDPG - Stat Tests

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1 Imports

```
[1]: import numpy as np
import pandas as pd
from scipy.stats import kruskal
import scikit_posthocs as sp
```

2 Logit, Firm 1

2.1 Data Loading

```
[2]: model = 'logit' # row number = 0

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
```

```

df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/ddpg_vs_ddpg.
↪csv")
df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/ddpg_vs_ddpg_schemeA.
↪csv")
df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/ddpg_vs_ddpg_schemeB.
↪csv")
df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/ddpg_vs_ddpg_schemeC.
↪csv")

# average prices
avg_prices["shock_none"].append(df_shock_none.iat[0, 1])
avg_prices["shock_a"].append(df_shock_a.iat[0, 1])
avg_prices["shock_b"].append(df_shock_b.iat[0, 1])
avg_prices["shock_c"].append(df_shock_c.iat[0, 1])

# deltas
deltas["shock_none"].append(df_shock_none.iat[0, 6])
deltas["shock_a"].append(df_shock_a.iat[0, 6])
deltas["shock_b"].append(df_shock_b.iat[0, 6])
deltas["shock_c"].append(df_shock_c.iat[0, 6])

# rpdis
rpdis["shock_none"].append(df_shock_none.iat[0, 8])
rpdis["shock_a"].append(df_shock_a.iat[0, 8])
rpdis["shock_b"].append(df_shock_b.iat[0, 8])
rpdis["shock_c"].append(df_shock_c.iat[0, 8])

```

2.2 Kruskal-Wallis Test - Average Prices

```

[3]: h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:
↪{epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on Average Prices

H-statistic: 16.36, P-value: 0.00095878, epsilon_squared: 0.6058

2.3 Post-Hoc Testing - Average Prices

```
[4]: dunn_matrix = sp.posthoc_dunn(
    [
        avg_prices["shock_none"],
        avg_prices["shock_a"],
        avg_prices["shock_b"],
        avg_prices["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

	1	2	3	4
1	1.000000	0.217928	0.655033	0.027669
2	0.217928	1.000000	0.032807	0.655033
3	0.655033	0.032807	1.000000	0.001749
4	0.027669	0.655033	0.001749	1.000000

- avg_prices["shock_none"] is significantly different from avg_prices["shock_c"]
- avg_prices["shock_a"] is significantly different from avg_prices["shock_b"]
- avg_prices["shock_b"] is significantly different from avg_prices["shock_c"]

2.4 Kruskal-Wallis Test - Deltas

```
[5]: h_stat, p_val = kruskal(
    deltas["shock_none"],
    deltas["shock_a"],
    deltas["shock_b"],
    deltas["shock_c"]
)

n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Deltas")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
```

Kruskal-Wallis Test on Deltas

H-statistic: 23.03, P-value: 0.00003976, epsilon_squared: 0.8531

2.5 Post-Hoc Testing - Deltas

```
[6]: dunn_matrix = sp.posthoc_dunn(
    [
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
    ],
```

```

        deltas["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)

```

	1	2	3	4
1	1.000000	0.022581	0.514901	0.000120
2	0.022581	1.000000	0.102911	0.268361
3	0.514901	0.102911	1.000000	0.001506
4	0.000120	0.268361	0.001506	1.000000

- `deltas["shock_none"]` is significantly different from `deltas["shock_a"]`
- `deltas["shock_none"]` is significantly different from `deltas["shock_c"]`
- `deltas["shock_b"]` is significantly different from `deltas["shock_c"]`

2.6 Kruskal-Wallis Test - RPDIs

```

[7]: h_stat, p_val = kruskal(
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on RPDIs")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on RPDIs

H-statistic: 23.07, P-value: 0.00003911, epsilon_squared: 0.8543

2.7 Post-Hoc Testing - RPDIs

```

[8]: dunn_matrix = sp.posthoc_dunn(
        [
            rpdis["shock_none"],
            rpdis["shock_a"],
            rpdis["shock_b"],
            rpdis["shock_c"]
        ], p_adjust='holm'
    )

    print(dunn_matrix)

```

	1	2	3	4
--	---	---	---	---

```

1  1.000000  0.033877  0.625869  0.000143
2  0.033877  1.000000  0.095807  0.222473
3  0.625869  0.095807  1.000000  0.000928
4  0.000143  0.222473  0.000928  1.000000

```

- `rpdis["shock_none"]` is significantly different from `rpdis["shock_a"]`
- `rpdis["shock_none"]` is significantly different from `rpdis["shock_c"]`
- `rpdis["shock_b"]` is significantly different from `rpdis["shock_c"]`

3 Hotelling, Firm 1

3.1 Data Loading

```

[9]: model = 'hotelling' # row number = 1

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/ddpg_vs_ddpg.
↳ csv")
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/ddpg_vs_ddpg_schemeA.
↳ csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/ddpg_vs_ddpg_schemeB.
↳ csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/ddpg_vs_ddpg_schemeC.
↳ csv")

```

```

# average prices
avg_prices["shock_none"].append(df_shock_none.iat[1, 1])
avg_prices["shock_a"].append(df_shock_a.iat[1, 1])
avg_prices["shock_b"].append(df_shock_b.iat[1, 1])
avg_prices["shock_c"].append(df_shock_c.iat[1, 1])

# deltas
deltas["shock_none"].append(df_shock_none.iat[1, 6])
deltas["shock_a"].append(df_shock_a.iat[1, 6])
deltas["shock_b"].append(df_shock_b.iat[1, 6])
deltas["shock_c"].append(df_shock_c.iat[1, 6])

# rpdis
rpdis["shock_none"].append(df_shock_none.iat[1, 8])
rpdis["shock_a"].append(df_shock_a.iat[1, 8])
rpdis["shock_b"].append(df_shock_b.iat[1, 8])
rpdis["shock_c"].append(df_shock_c.iat[1, 8])

```

3.2 Kruskal-Wallis Test - Average Prices

```

[10]: h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on Average Prices

H-statistic: 8.76, P-value: 0.03268365, epsilon_squared: 0.3244

3.3 Post-Hoc Testing - Average Prices

```

[11]: dunn_matrix = sp.posthoc_dunn(
    [
        avg_prices["shock_none"],
        avg_prices["shock_a"],
        avg_prices["shock_b"],
        avg_prices["shock_c"]
    ], p_adjust='holm'
)

```

```
print(dunn_matrix)
```

	1	2	3	4
1	1.000000	0.128587	1.000000	0.255254
2	0.128587	1.000000	0.128587	1.000000
3	1.000000	0.128587	1.000000	0.255254
4	0.255254	1.000000	0.255254	1.000000

As none of the entries of the Dunn matrix is significant (< 0.05), there is no significant difference between any of the average prices.

3.4 Kruskal-Wallis Test - Deltas

```
[12]: h_stat, p_val = kruskal(
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on Deltas")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
```

Kruskal-Wallis Test on Deltas

H-statistic: 10.34, P-value: 0.01586183, epsilon_squared: 0.3831

3.5 Post-Hoc Testing - Deltas

```
[13]: dunn_matrix = sp.posthoc_dunn(
        [
            deltas["shock_none"],
            deltas["shock_a"],
            deltas["shock_b"],
            deltas["shock_c"]
        ], p_adjust='holm'
    )

    print(dunn_matrix)
```

	1	2	3	4
1	1.000000	0.117792	1.000000	0.079714
2	0.117792	1.000000	0.120366	1.000000
3	1.000000	0.120366	1.000000	0.117792
4	0.079714	1.000000	0.117792	1.000000

As none of the entries of the Dunn matrix is significant (< 0.05), there is no significant difference between any of the average prices.

3.6 Kruskal-Wallis Test - RPDIs

```
[14]: h_stat, p_val = kruskal(
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on RPDIs")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
```

Kruskal-Wallis Test on RPDIs

H-statistic: 3.18, P-value: 0.36429349, epsilon_squared: 0.1179

There are no significant differences between any of the RPDIs (P-value is not significant).

3.7 Post-Hoc Testing - RPDIs

Only performed if the Kruskal-Wallis P-value is significant.

4 Linear, Firm 1

4.1 Data Loading

```
[15]: model = 'linear' # row number = 2

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
```



```

    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/ddpg_vs_ddpg.
↪csv")
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/ddpg_vs_ddpg_schemeA.
↪csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/ddpg_vs_ddpg_schemeB.
↪csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/ddpg_vs_ddpg_schemeC.
↪csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[2, 1])
    avg_prices["shock_a"].append(df_shock_a.iat[2, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[2, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[2, 1])

    # deltas
    deltas["shock_none"].append(df_shock_none.iat[2, 6])
    deltas["shock_a"].append(df_shock_a.iat[2, 6])
    deltas["shock_b"].append(df_shock_b.iat[2, 6])
    deltas["shock_c"].append(df_shock_c.iat[2, 6])

    # rpdis
    rpdis["shock_none"].append(df_shock_none.iat[2, 8])
    rpdis["shock_a"].append(df_shock_a.iat[2, 8])
    rpdis["shock_b"].append(df_shock_b.iat[2, 8])
    rpdis["shock_c"].append(df_shock_c.iat[2, 8])

```

4.2 Kruskal-Wallis Test - Average Prices

```

[16]: try:
    h_stat, p_val = kruskal(
        avg_prices["shock_none"],
        avg_prices["shock_a"],
        avg_prices["shock_b"],
        avg_prices["shock_c"]
    )

    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

```

```

    print("Kruskal-Wallis Test on Average Prices")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
except ValueError:
    print("All numbers are identical in kruskal")

    print(f"avg_prices[\"shock_none\"] = {[float(x) for x in avg_prices[\"shock_none\"]]}")
    print(f"avg_prices[\"shock_a\"] = {[float(x) for x in avg_prices[\"shock_a\"]]}")
    print(f"avg_prices[\"shock_b\"] = {[float(x) for x in avg_prices[\"shock_b\"]]}")
    print(f"avg_prices[\"shock_c\"] = {[float(x) for x in avg_prices[\"shock_c\"]]}")

```

Kruskal-Wallis Test on Average Prices

H-statistic: 3.74, P-value: 0.29112569, epsilon_squared: 0.1385

There are no differences between any of the average prices.

4.3 Post-Hoc Testing - Average Prices

Only performed if the Kruskal-Wallis P-value is significant.

4.4 Kruskal-Wallis Test - Deltas

```

[17]: h_stat, p_val = kruskal(
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
    )
    n = 4 * 7 # total number of observations across all groups
    epsilon_squared = h_stat / (n - 1)

    print("Kruskal-Wallis Test on Deltas")
    print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")

```

Kruskal-Wallis Test on Deltas

H-statistic: 22.92, P-value: 0.00004202, epsilon_squared: 0.8488

4.5 Post-Hoc Testing - Deltas

```
[18]: dunn_matrix = sp.posthoc_dunn(
      [
          deltas["shock_none"],
          deltas["shock_a"],
          deltas["shock_b"],
          deltas["shock_c"]
      ], p_adjust='holm'
    )

print(dunn_matrix)
```

	1	2	3	4
1	1.000000	0.066739	1.000000	0.000398
2	0.066739	1.000000	0.066739	0.221102
3	1.000000	0.066739	1.000000	0.000398
4	0.000398	0.221102	0.000398	1.000000

- deltas["shock_none"] is significantly different from deltas["shock_c"]
- deltas["shock_b"] is significantly different from deltas["shock_c"]

4.6 Kruskal-Wallis Test - RPDIs

```
[19]: h_stat, p_val = kruskal(
      rpd["shock_none"],
      rpd["shock_a"],
      rpd["shock_b"],
      rpd["shock_c"]
    )

n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on RPDIs")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared: {epsilon_squared:.4f}\n")
```

Kruskal-Wallis Test on RPDIs

H-statistic: 4.86, P-value: 0.18244897, epsilon_squared: 0.1799

There are no differences between any of the RPDIs.

4.7 Post-Hoc Testing - RPDIs

Only performed if the Kruskal-Wallis P-value is significant.